



A Level • Cambridge (CIE) • Chemistry

34 mins

18 questions

Multiple Choice Practice

Topical Practice MCQ

21. Organic synthesis

Carefully selected past paper questions by topic.

32 Propanoic acid can be made from bromoethane using a two-stage synthesis.

Which pair of reagents is most suitable?

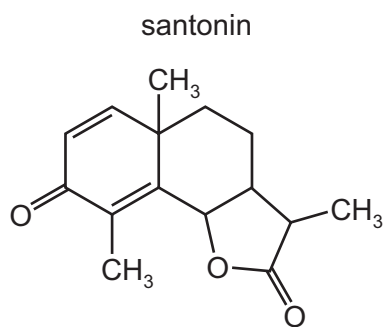
	reagent for stage 1	reagent for stage 2
A	hydrogen cyanide	aqueous sodium hydroxide
B	aqueous sodium hydroxide	excess acidified potassium dichromate(VI)
C	ethanolic sodium hydroxide	acidified potassium manganate(VII)
D	potassium cyanide	dilute hydrochloric acid

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29 The structure of santonin is shown.



Santonin is first treated with warm dilute H_2SO_4 . The product of this reaction is treated with cold dilute acidified KMnO_4 . A final product, Q, is obtained.

How many atoms of hydrogen in each molecule of product Q will react with sodium metal?

A 2

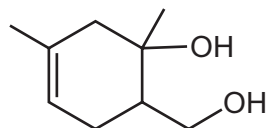
B 4

C 5

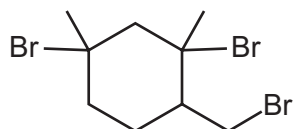
D 6

29 What is the major product formed when compound R is warmed with an excess of HBr?

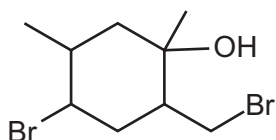
R



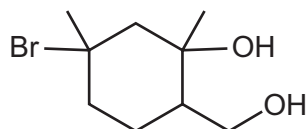
A



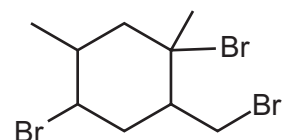
B



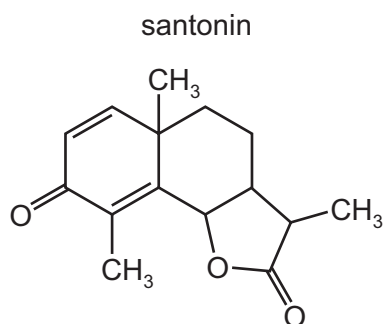
C



D



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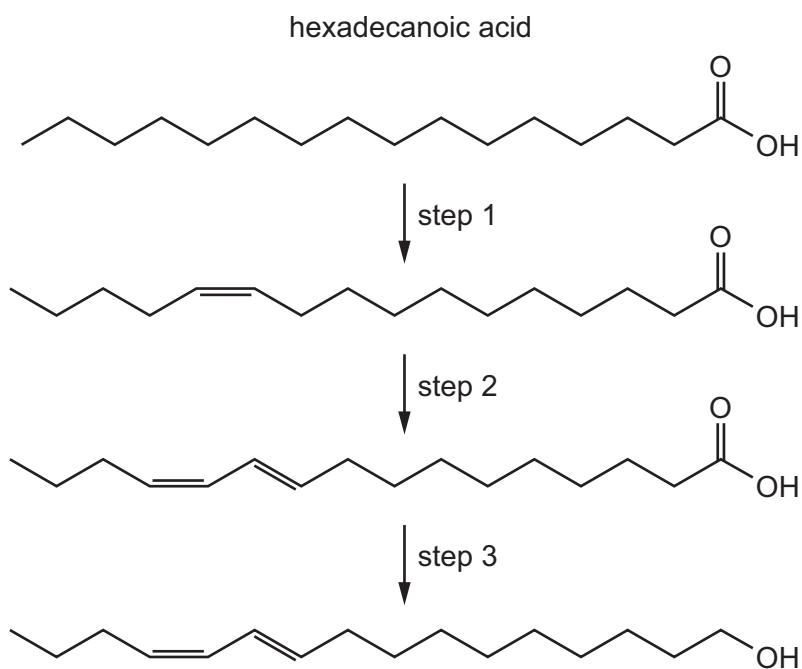
A 2

B 4

C 5

D 6

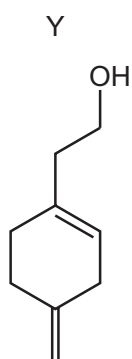
- 21 Hexadeca-10,12-dien-1-ol is produced by silk moths from hexadecanoic acid in a three-step enzymic process.



Which row contains correct descriptions of the three steps?

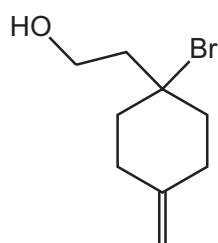
	step 1	step 2	step 3
A	elimination	elimination	dehydration
B	elimination	reduction	reduction
C	oxidation	elimination	oxidation
D	oxidation	oxidation	reduction

26 An excess of dry HBr is warmed with compound Y.

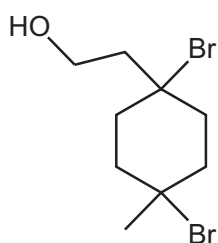


What is the major product of the reaction?

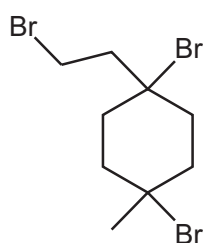
A



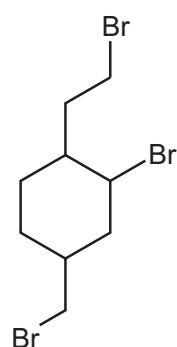
B



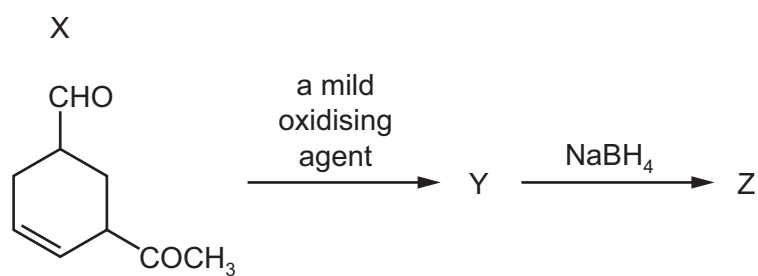
C



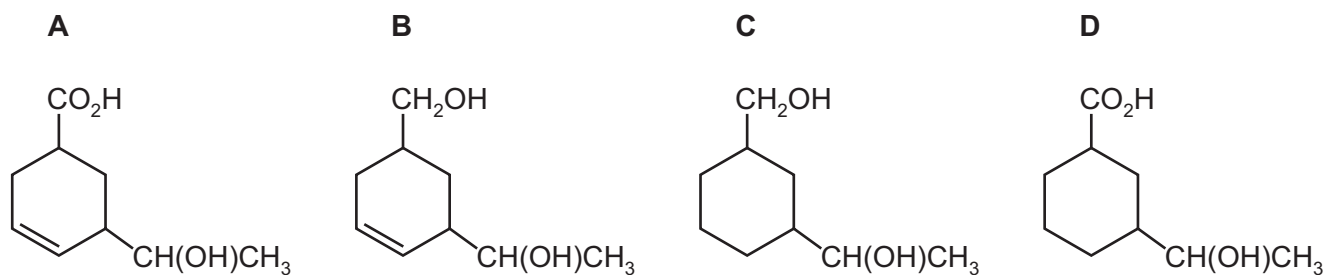
D



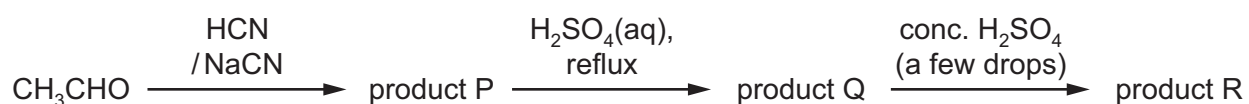
28 Compound X is treated with two reagents successively, forming compound Z.



What could be Z?



28 Ethanal, CH_3CHO , is used to make product R in a three-stage synthesis.



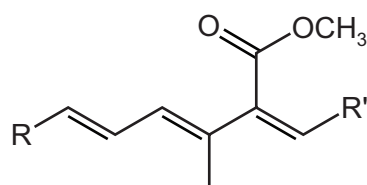
Two molecules of Q react to give one molecule of R plus two molecules of water.

R has two ester functional groups in each molecule. R does not react with sodium.

What is the empirical formula of R?

- A** CHO **B** $\text{C}_3\text{H}_4\text{O}_2$ **C** $\text{C}_3\text{H}_5\text{O}_2$ **D** $\text{C}_6\text{H}_{10}\text{O}_5$

23 Part of the structure of a fungicide, strobilurin, is shown. R and R' are inert groups.

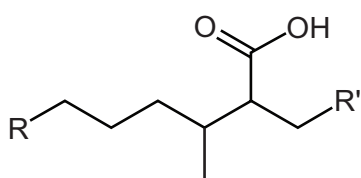


strobilurin

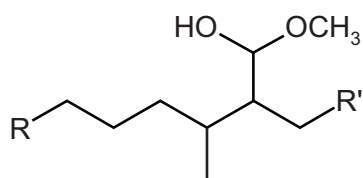
In this reaction, strobilurin is warmed with aqueous sulfuric acid producing compound X. Compound X is then treated with hydrogen in the presence of a nickel catalyst producing compound Y.

What could be the structure of compound Y?

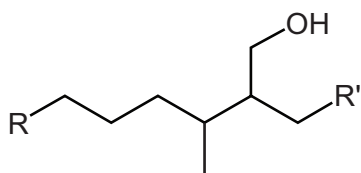
A



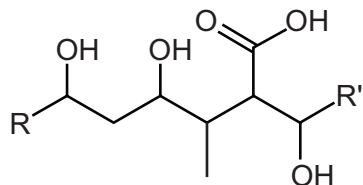
B



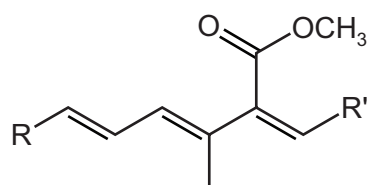
C



D



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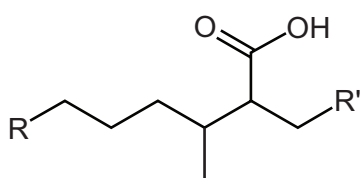


strobilurin

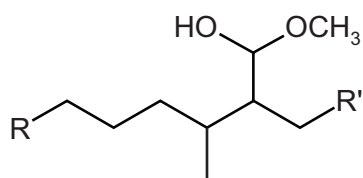
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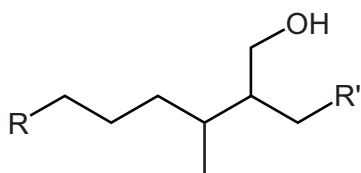
A



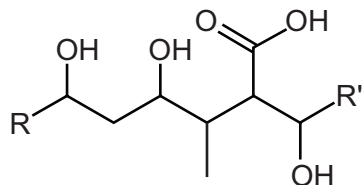
B



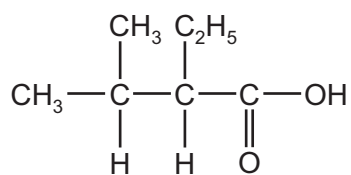
C



D

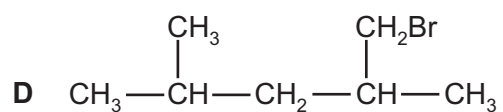
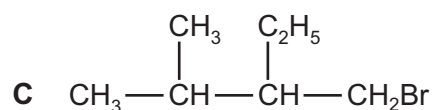
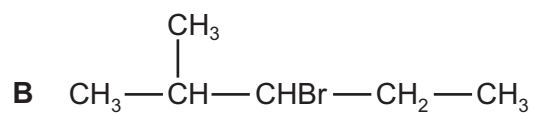
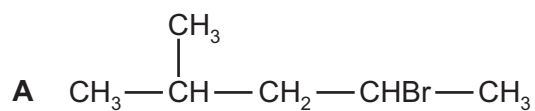


27 The compound 2-ethyl-3-methylbutanoic acid is used as a flavouring in some food.

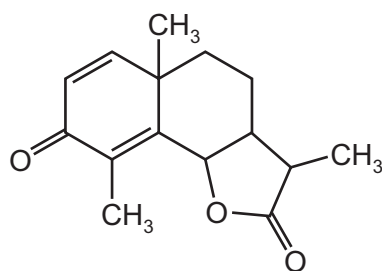


2-ethyl-3-methylbutanoic acid

Which compound will produce 2-ethyl-3-methylbutanoic acid when heated under reflux with sodium cyanide in ethanol, followed by acid hydrolysis of the reaction product?



27 Santonin is a drug that was once widely used to expel parasitic worms from the body.



santonin

When santonin is first treated with warm dilute H_2SO_4 , and then the product of this reaction is treated with cold acidified KMnO_4 , a final product **X** is obtained.

How many atoms of hydrogen in each molecule of product **X** can be displaced with sodium metal?

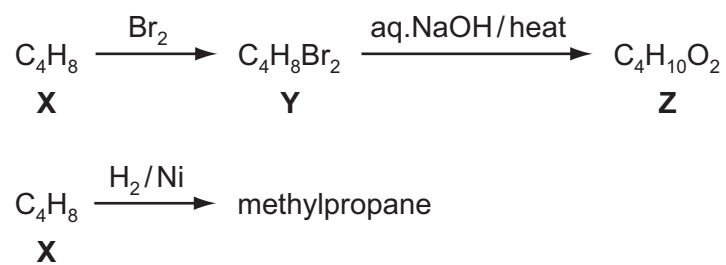
A 2

B 4

C 5

D 6

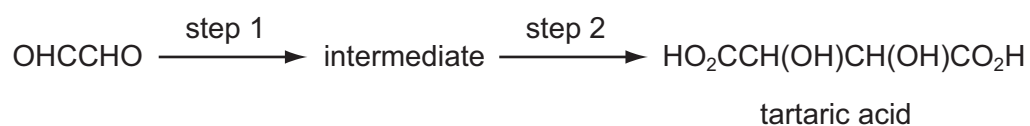
29 Compound X, molecular formula C_4H_8 , undergoes the following reactions.



What is the formula of compound Z?

- A $CH_3CH_2CH(OH)CH_2OH$
- B $CH_3CH(OH)CH(OH)CH_3$
- C $CH_3CH(CH_2OH)CH_2OH$
- D $(CH_3)_2C(OH)CH_2OH$

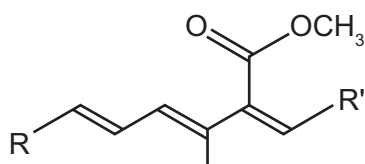
23 Tartaric acid is present in some wines. It may be synthesised in the laboratory in two steps.



Which reagents could be used for this synthesis?

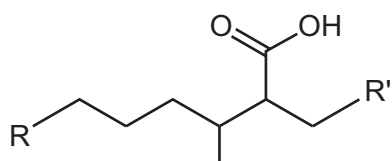
	step 1	step 2
A	HCl(aq)	HCN(g)
B	HCN, NaCN(aq/alcoholic)	H ₂ SO ₄ (aq)
C	H ₂ SO ₄ (aq)	K ₂ Cr ₂ O ₇ /H ₂ SO ₄ (aq)
D	KCN(aq/alcoholic)	K ₂ Cr ₂ O ₇ /H ₂ SO ₄ (aq)

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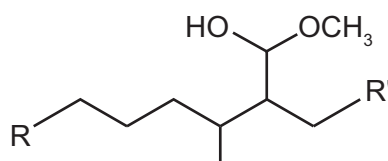


If strobilurin is first warmed with aqueous sulfuric acid, and its product then treated with hydrogen in the presence of a palladium catalyst, what could be the structure of the final product?

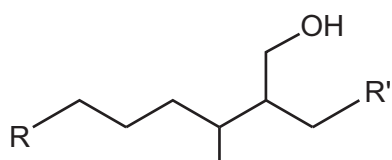
A



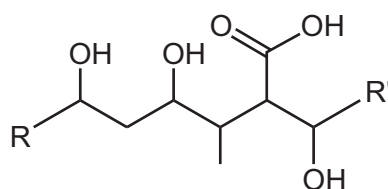
B



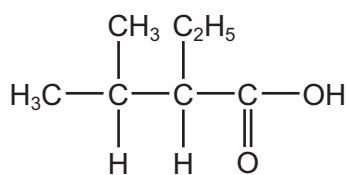
C



D

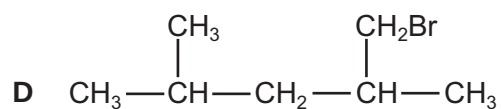
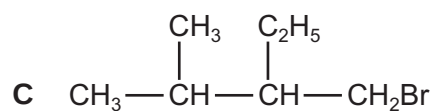
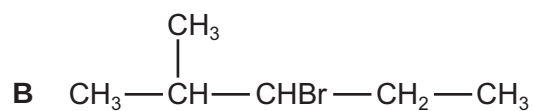
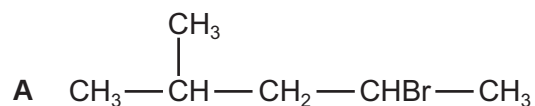


- 29 The characteristic odour of rum is attributed to the compound 2-ethyl-3-methylbutanoic acid.

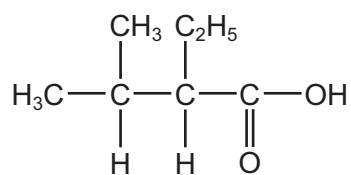


2-ethyl-3-methylbutanoic acid

Which compound will produce 2-ethyl-3-methylbutanoic acid by heating under reflux with alcoholic sodium cyanide and subsequent acid hydrolysis of the reaction product?

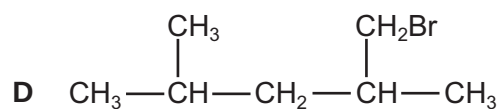
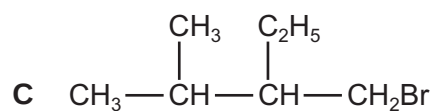
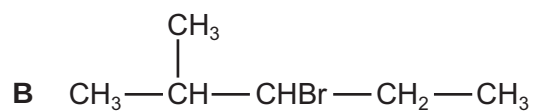
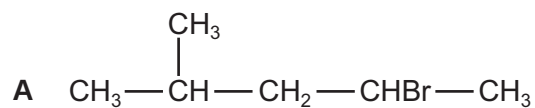


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Answer Key

No.	Paper	Topic	Answer
1	9701_w23_qp_11	21. Organic synthesis	D
2	9701_w23_qp_13	21. Organic synthesis	D
3	9701_w22_qp_11	21. Organic synthesis	D
4	9701_w22_qp_12	21. Organic synthesis	A
5	9701_w22_qp_13	21. Organic synthesis	D
6	9701_s21_qp_12	21. Organic synthesis	D
7	9701_m20_qp_12	21. Organic synthesis	C
8	9701_s19_qp_11	21. Organic synthesis	A
9	9701_s18_qp_12	21. Organic synthesis	B
10	9701_w16_qp_11	21. Organic synthesis	A
11	9701_w16_qp_13	21. Organic synthesis	A
12	9701_w14_qp_13	21. Organic synthesis	B
13	9701_w12_qp_13	21. Organic synthesis	D
14	9701_w11_qp_12	21. Organic synthesis	D
15	9701_w10_qp_11	21. Organic synthesis	B
16	9701_w10_qp_12	21. Organic synthesis	A
17	9701_w09_qp_11	21. Organic synthesis	B
18	9701_w09_qp_12	21. Organic synthesis	B